

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-a781891dc0ad787e7.jsonl`  
- SHA-256: `e36ddddd1234fa4722d9de6fda0d277689d921f15273bfe5fd683a312cf931277`  
- Source modified: `2026-06-19T17:19:52+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_4e32c8b8-725`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T17:18:10.920Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:

- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): ANALYZERS_AND_RECONCILERS.md, PERSONALIZATION_PLAN.md, PER_VIDEO_OUTPUT_LAYOUT.md, DESKTOP_APP_PLAN.md, I18N_PLAN.md, EXPERIMENT_HARNESS.md, ABLATION_LAYER.md

For EACH doc:

1. Read the status header + TL;DR/intro + (if present) the \$7/\$8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.
5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:13.310Z)

I'll audit each doc in my batch. Let me start by reading them and gathering cross-reference evidence in parallel.

3. user (2026-06-19T17:18:17.124Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end' from the remote, but no such ref was fetched.

```
---ARCHIVES---
CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
COMMERCIAL_READINESS_REVIEW.md
DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
MONETIZATION_AUDIT.md
README.md
TEST_RUN_SUMMARY.md
checkpoints
decisions
past_projects
quality-iterations
research
---ARCHIVES-SUBDIRS---
docs/archives/checkpoints/:
2026-06-14-hardening-loop-complete.md
2026-06-eval-baseline
2026-06-logging-reporting
2026-06-scaffolding-complete
2026-06-strategy-foundation

docs/archives/decisions/:
DECISIONS.md

docs/archives/past_projects/:
README.md
low-regret-moves

docs/archives/quality-iterations/:
2026-06-18-lovo-resweep-n13
2026-06-first-ab-experiment
2026-06-human-gt-tuning
2026-06-initial
2026-06-multivideo-lovo
README.md

docs/archives/research/:
AUDIO_E1_FINDINGS.md
AUDIO_RALLY_DETECTION_PLAN.md
OWNED_MODEL_TRAINING_STUDY.md
RALLY_DETECTION_RESEARCH.md
```

4. user (2026-06-19T17:18:17.145Z)

```
1      # Personalization Plan ? a per-user feature on a generalization-first baseline
2
3      **Status: DESIGN ? owner-adopted (hybrid); Phase-0 engineering landing.** This is the
```

authoritative

4 plan for the personalization track. The detailed strategy analysis + the full pros/cons that led here

5 live in the working artifacts `scratch/PERSONALIZATION\_PIVOT\_ANALYSIS.md` (the decision-grade

6 counter-case) and `scratch/PERSONALIZATION\_FEATURE\_DESIGN.md` (the code-grounded design); this doc is

7 the checked-in summary + decisions + phased build plan. Peer to

8 OWNED_MODEL_IMPLEMENTATION_PLAN.md,

9 EXPERIMENT_HARNESS.md, and

MULTI_SIGNAL_FUSION_PLAN.md.

10

11 ## 0. Thesis & strategic rationale (why *hybrid*, not personalization-first)

12

13 **Foundation principle (owner):** "a tool that gets better the more you help it."

Contribution

14 (data / video / diagnostics / annotation) improves BOTH the user's own model AND the shared baseline;

15 **payoff scales with contribution.**

16

17 A directional pivot ? *personalization-first* (ship a baseline that continually self-optimizes to one

18 user's own footage) ? was researched in depth. The verdict was **HYBRID, not all-in**:

19

20 - **Adopt personalization as a flag-gated FEATURE on a generalization-first baseline.**

The harness

21 genuinely makes per-user *tuning* near-free (it is already a corpus-relative A/B engine; a per-user

22 corpus is just a special case). This is real and worth shipping.

23 - **Do NOT bet the strategy on per-user *learning** Per-user A/B is the n?6 small-sample regime

24 *worse* (sharding data when you should pool); the differentiated parts (a learned per-user model

25 serving path, the correction?label write path, a user/tenant dimension) are largely unbuilt; a

26 naively-overfit per-user model loses the cross-user data flywheel and has no shared baseline to

27 regress against.

28 - **The owner's contribution-flywheel + free-tier-pooling closes the one real hole** the counter-case

29 found: by having (free-tier) users contribute to the *shared* baseline, the compounding cross-user

30 moat is preserved *alongside* personalization ? personalization **and** flywheel, not either/or.

31

32 Net: a generalization-first **baseline** ("batteries"), a thin **per-user tuning** layer gated by the

33 honest small-N statistics (with a hard floor + structural-guard protection), and a **contribution

34 loop** that feeds both.

35

36 ## 1. Owner decisions (locked 2026-06-14)

37

38 1. **Identity** ? auth WILL land; *design for the north star (multi-user / cloud), implement for

39 current use (single-user local)*. `user\_id` nullable now (NULL = local install), device/install

40 UUID for the alpha; real auth + a `users` table later.

41 2. **Per-user model storage** ? inline `classifier\_json` in the (local) DB now, but **behind a load

42 interface** so that post-auth it is fetched from the **cloud-hosted DB run by the service** that

43 generates base models and orchestrates the product. The local store is one implementation of that

44 interface.

```

45     3. `min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate.** A
brand-new user with
46     one video is ALWAYS fully segmented by the baseline batteries ? never denied, never
"give me more
47     videos first." `min_n` only gates whether a *per-user* A/B is trusted enough to
OVERRIDE a baseline
48     cue default *for that user*; below it (or if the CI / sign test don't clear) ? keep
the baseline.
49     **Training-corpus policy:** *no more than needed, no less than all-if-few* ? fit on
the
50     `smallest_k_with_lift` videos (the held-out-USER learning curve), else ALL available.
51     4. ***"Flip Gate-A to default"*** = ship the validated Gate-A held-shuttle guard as the
**default
52     detection path** (today it lives in the opt-in `FusionSegmenter`, which is not the
default
53     segmenter). Changes served behavior ? **gated on a golden-set regression run + owner
OK**; still
54     open (see §5).
55     5. **Flywheel = backend property** (not in-product marketing). A **free tier** is likely
and
56     **free-tier videos ARE pooled** into the baseline (consent baked into the free terms;
57     minors-exclusion still enforced) ? the clean value-exchange that feeds the owner-
trained base model.
58     Paid tiers get stronger privacy / opt-out.
59
60     **Pending sub-decision:** structural-guard scope ? recommended **Gate-A only** (the
proven structural

```

5. user (2026-06-19T17:18:17.154Z)

```

1     # A signal-fusion framework for rally detection ? analyzers, scopes & reconcilers
2
3     **Status:** **DESIGN ? owner-gated, NOT started.** A peer to
4     [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) and
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md).
5     This doc specs a small **framework** for evolving rally-detection quality: producers
that emit **signals**
6     at three temporal **scopes** (frame / window / session), a **scorer** that *learns to
weight* them, and an
7     auditable **reconciliation** pass that lints the decision against the evidence. The goal
is to keep piling
8     on clever cues **without the pipeline becoming a pile of special cases.** Docs-only; no
implementation
9     code. Each phase below is a separate, owner-gated PR.
10
11     > **Companion docs:** the detection-layer producers (shuttle / person / audio / shuttle-
state) live in
12     > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) (`RallyCue` = swappable
producer; the
13     > `_enrich_candidate_stats` hook; scale/translation-invariant feature discipline, §3.2);
the no-regression
14     > A/B + LOVO measurement these ride is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)
(`Variant`, `compare`,
15     > `compare_unlabeled`); the scorer is the distilled logistic regression (**ADR-004**,
16     > `backend/eval/classifier.py`); the modular rally layer is **ADR-007**
17     > ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the 2-layer
mental model is
18     > [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md); the live quality log is
19     > [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md).
20
21     > **Framing.** Rally detection is a genuinely hard temporal problem ? weak, multi-modal
cues; interval-level
22     > ground truth that is scarce and expensive; a confound (warm-up / knock-ups / multi-
court / held-shuttle)
23     > that looks identical to real play in any single frame or frequency band. This doc is

```

```

written as a
24     > **reusable framework**, not a badminton-specific patch: the contracts are sport-
agnostic and the design
25     > is meant to generalize to other weakly-supervised temporal-segmentation problems
($10). Where that
26     > ambition touches research claims, $10 says plainly what is and isn't yet
substantiated.
27
28     ---
29
30     ## 1. The spine ? NO special-casing in the decision path
31
32     This is the one principle everything else serves:
33
34     > **Every producer (analyzer, detector, reconciler) emits a SIGNAL carrying a value + a
confidence/presence.
35     > The SCORING MODEL consumes those signals and *learns to weight them*. We do NOT add
hand-coded `if/else`
36     > branches into the decision.**
37
38     **Producers propose; the scorer arbitrates; the experiment harness + golden set decide
what gets promoted.**
39     That separation is what lets us add a net-hit service fault, a let-cord, a double-
bounce, a shuttle-out
40     cue, a *warm-up span*, a *shuttle-state* ? each as "register a producer + a feature
column," never **add
41     a branch.** A signal that doesn't earn its keep stays at weight ? 0; it costs nothing
because it is an
42     unused column, not a live `if`.
43
44     The pattern already exists in miniature: `experiment.predict()`
(`backend/eval/experiment.py:213`) treats
45     `net_crossings` and `players_in_court` as **feature columns** ? `_gate_mask` is the
trivial scorer
46     (threshold a column), `LogisticRegression` (ADR-004) the richer learned scorer (weight
all columns), a
47     future scorer richer still. The decision path is *signals ? scorer*, with **zero** `if
service_fault`:
48     drop.. Analyzers feed it more columns; reconcilers run *outside* it as an audited post-
pass.
49
50     ...
51     DETECTION ? frame-scope producers (expensive, GPU/WSL, run ONCE)
52     WASB shuttle (x,y,v,vis) · person boxes+tracks · audio hits · shuttle-
state(in_air/contact/inactive)*
53                                     ? per-frame-aligned signals (+ net estimate, window
bounds)
54                                     ?
55     shuttle-seeded candidate windows    (FusionSegmenter, P2)
56                                     ?
57     ANALYZERS ? decision-layer producers (pure, offline, re-scorable)           [forward
seam]
58     ?? window-scope (WindowAnalyzer): one candidate ? features + events + boundary
hints e.g. service-fault
59     ?? session-scope (SessionAnalyzer): whole timeline ? labeled spans + per-window
membership e.g. warm-up
60                                     ? every signal ? candidate_segments.stats as feature
columns

```

6. user (2026-06-19T17:18:17.659Z)

```

1     # Per-video output layout ? one directory per video (design / plan)
2
3     **Status:** **DESIGN ? owner-gated, NOT started.** Proposed convention so processing N
videos

```

```

4     yields N self-contained subdirectories instead of one tangled `output/`. Docs-only.
5
6     ## The problem
7     Today every run writes into a **flat `output/`** with filename prefixes ? highlights
8     (`<video>_highlights.mp4`), run logs (`<video>_run.log`), eval CSVs
9     (`<video>_vs_human.csv`),
10    the shared SQLite DB (`sports_indexer.db`), plus **hardcoded shared subdirs** the CLI
11    `--output-dir` does **not** redirect:
12    - `output/wasb/` and `output/tracknet/` ? trajectory CSVs + frame caches
13    (`trajectory_hybrid.py:~176`, `out_dir = os.path.join("output", self.family)` ?
14    literal `"output"`).
15    - `output/chunks_<provider>_handoff/` ? the AI-handoff clip staging (same file).
16
17    So 5 videos produce an interleaved pile, and `--output-dir` only relocates *some*
18    artifacts
19    (highlights/clips/DB) while trajectories and handoff clips still land in the shared
20    `output/`.
21
22    The first isolated run (`output/GX010125_run/`, 2026-06-14) had to be done by hand and
23    is only
24    *partially* isolated for this reason.
25
26    ## The target
27    ```
28    output/
29    <video_slug>/                                # slug = sanitized stem + short video_id (collision-
30    safe)
31    runlog.txt                                    # the per-video run log (today: output/<video>_run.log)
32    sports_indexer.db                             # per-video DB (or a row-scoped view of a shared DB ?
33    see below)
34    wasb/<key>_wasb.csv                            # the trajectory (today: output/wasb/)
35    handoff/                                       # AI-handoff clips (today: output/chunks_*_handoff/,
36    auto-cleaned)
37    rallies.csv                                  # detected rally [start,end] + shot_count
38    segments/                                    # per-segment clips / pngs / debug frames
39    highlights.mp4                                # the reel
40    report.json / report.md                       # the observability report
41    ```
42
43    A directory with 5 videos ? 5 subdirectories, each holding **everything** for that
44    video.
45
46    ## Design
47    - **`video_slug(video_path, video_id) -> str`** ? a new helper (next to
48    `utils/hashing.py`):
49    sanitized stem + `video_id[:12]` (the MD5 already used as the cache key), e.g.
50    `GX010125_1080p__89605e80ed01`. Collision-safe (same stem / different bytes ?
51    different dir).
52    - **Thread one `run_dir` through the pipeline** instead of the literal `"output"`:
53    - `TrajectoryHybridSegmenter.process_video` ? `out_dir = os.path.join(run_dir,
54    self.family)`
55    and the handoff dir under `run_dir` (replaces the two hardcoded `"output"` joins).
56    - The `wasb_runner`/`tracknet_runner` already take `output_win_dir` ? just pass the
57    per-video one.
58    - The CLI `--output-dir` (already in `backend/main.py`) becomes the **parent** ; the
59    pipeline
60    creates `<output-dir>/<video_slug>/` and writes everything under it.
61    - **DB choice (owner decision):** (a) a per-video `sports_indexer.db` in the slug dir
62    (simplest,
63    fully isolated, but loses cross-video queries), or (b) keep the shared DB and only
64    isolate
65    *files* (the slug dir holds artifacts; the DB stays one queryable store keyed by
66    `video_id`).
67    **Recommend (b)** ? the DB is already `video_id`-scoped; isolate the heavy *file*
68    artifacts,
69    keep one queryable metadata store. The slug dir then omits its own DB.
70    - **Back-compat:** default parent stays `output/`; a flag (or just the default-on

```

```

behaviour)
51     switches flat?per-video. Old flat outputs are left untouched; new runs go to slug
dirs.
52
53     ## Phasing (owner-gated; ONE PR per slice, off `master`)
54     - **L0 ? this plan.**
55     - **L1 ? `video_slug` helper + thread `run_dir`** through `trajectory_hybrid` (replace
the two
56     hardcoded `"output"` joins) + the per-video subdir creation; CLI `--output-dir` =
parent.
57     Pure plumbing; a regression test pinning that all artifacts land under the slug dir.
58     - **L2 ? runlog + report + rallies.csv into the slug dir** ; migrate the eval/CSV
writers.
59     - **L3 ? cache-hygiene + cleanup_caches awareness** of the new layout (per-video purge).
60

```

7. assistant (2026-06-19T17:18:19.857Z)

Let me continue reading the remaining docs and gather cross-reference data.

8. user (2026-06-19T17:18:20.809Z)

```

1     # Cross-Platform Desktop App ? "Plug in your card, get your highlights"
2
3     > **Status:** `DRAFT` ? `owner-gated` (scoping doc only; **nothing is built**) .
**Owner:** avidullu . **Created:** 2026-06-18 . **Last updated:** 2026-06-18
4     > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once the owner
greenlights P0).
6     > **Relation to existing docs:** **peer-of**
[ `PLATFORM_ARCHITECTURE.md` ] (PLATFORM_ARCHITECTURE.md) (this is the **`LocalDesktop`
ComputeBackend** made into a product) . **realizes the L0 tier** of
[ `VIDEO_LOCALITY_MODEL.md` ] (VIDEO_LOCALITY_MODEL.md) . **consumes**
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ] (OWNED_MODEL_IMPLEMENTATION_PLAN.md) (the smaller/export-
clean model is one answer to the compute crux) . bound by
[ `COMMERCIALIZATION.md` ] (COMMERCIALIZATION.md) licensing guardrails.
7     > **Honesty note:** every claim is tagged `[verified]` (checked against code/measured),
`[researched]` (needs sign-off), or `[design]` (proposed). **The entire build is `[design]`** ?
this doc scopes; it does not ship. Not legal advice.
8
9     ---
10
11     ## 0. TL;DR
12     Package the mature local pipeline (WASB shuttle detector + improved windowing + ffmpeg
stitch, already runnable fully-offline via [ `tools/generate_highlights.py --skip-ai-
handoff` ] (./tools/generate_highlights.py)) as a **cross-platform desktop app** for **non-
technical recreational players**: plug in the camera's USB/SD card, click once, get back **only
the highlight reels** ? fully on-device, **no upload**, **no copy of the GBs of 4K originals**.
This is the product surface that fits the "local, cheap, efficient" thesis and the L0 ("nothing
leaves the machine") tier of `VIDEO_LOCALITY_MODEL.md`.
13
14     **The make-or-break unknown is compute on a typical enthusiast laptop** ? which often
has **no discrete GPU**. WASB runs at **~3.4x real-time on a laptop RTX 2070 Super** `[verified,
measured]`; on CPU it is far slower, and Apple Silicon has **no CUDA** at all. **P0 is a
feasibility spike that measures CPU and Apple-MPS throughput for a ~10-min clip before any app
is built.** If CPU-only is unacceptable on commodity hardware, the path needs a smaller/faster
owned model (ties `OWNED_MODEL_IMPLEMENTATION_PLAN`) or a tiered "fast-if-GPU, slow-but-works-
if-CPU" design. **Everything past P0 is gated on that result.**
15
16     ## 1. Problem & goal
17
18     **Why now / the user.** Target users are **recreational sports enthusiasts, not tech-
savvy**. Their match videos are **huge** (GBs of 4K). The cloud-upload product surface fails

```

them on every axis: slow on a typical Indian uplink (`VIDEO\_LOCALITY\_MODEL.md` \$0: 10 GB ? ~1 h at 20 Mbps, ~4.5 h at 5 Mbps), bandwidth-costly, privacy-fraught, and it bloats both our storage and their drives. The thing the customer actually wants back is ****kilobytes of timestamps ? a highlight reel**** (`VIDEO\_LOCALITY\_MODEL.md` \$1: ****timestamps are the product; the video is dead weight****).

19
20 ****The concrete outcome ("good" looks like this):****
21 1. User plugs the GoPro/phone SD card or camera USB into their laptop.
22 2. The app detects the card, lists the match videos on it.
23 3. User clicks ****Generate Highlights****.
24 4. The app makes an on-device proxy, runs WASB detection + windowing + stitches the reel ? ****100% locally****.
25 5. The reel(s) land in a user-chosen folder. ****The originals are never copied off the card and never bloat any drive.****

26
27 No account, no upload, no cloud dependency in the default path. Gemini stays an ****optional**** "better brain if you want it" toggle ? never required, never bundled as a hard dependency, never a training source (`COMMERCIALIZATION` C4).

28
29 ****Read to get current:**** `VIDEO\_LOCALITY\_MODEL.md` (the L0/locality framing + OQ5 packaging seed), `PLATFORM\_ARCHITECTURE.md` \$3.2 (the ComputeBackend seam), [backend/pipeline/detectors/base.py](../backend/pipeline/detectors/base.py) (the detector ABC that already anticipates a Linux-native runner), `tools/generate\_highlights.py` (the local entrypoint this app wraps).

30
31 **## 2. Decisions locked**

32
33 | # | Decision | Source / date | Implication |
34 |---|-----|-----|-----|
35 | D1 | ****Default path is 100% on-device, no upload, no Gemini.**** The reel is produced from the local pipeline (`--skip-ai-handoff`). | This doc · 2026-06-18; matches `VIDEO\_LOCALITY\_MODEL` L0 | The app must never ***require*** a network call. Gemini is an opt-in toggle only. |
36 | D2 | ****Never copy the originals.**** The app reads source video in place (off the card / a chosen folder) and writes ****only**** reels + a tiny report to a user folder. | This doc; mirrors `generate\_highlights.py` non-destructive `\_atomic\_publish` discipline | No "import library" that duplicates 10 GB files. Detection runs on a proxy; stitch reads the original in place. |
37 | D3 | ****Drop the WSL2 dependency; run the detector natively per-OS.**** | This doc; enabled by the `DetectorRunner` ABC [verified] | Windows-native (no WSL), Linux+NVIDIA (CUDA directly), macOS (MPS or CPU). The WSL adapter stays for the current dev box but is ****not**** the shipped path. |
38 | D4 | ****Commercial-clean licensing for everything bundled: MIT / Apache-2.0 / BSD only**** (code, weights). ffmpeg ships as an ****LGPL build**** (no GPL-only encoders like x264). | ****COMMERCIALIZATION**** (ratified 2026-06-09); ****VIDEO_LOCALITY_MODEL**** OQ5 (****ffmpeg LGPL build = licensing-clean****) | WASB-SBDT weights are MIT [verified] (see `tracknet-wasb-wsl2-decision`). The stitch must use openh264 (BSD) or OS hardware encoders, ****not**** statically-linked x264. See \$3.4. |

39 | D5 | ****Reuse the existing pipeline verbatim; add only the shell + a native runner + device-I/O.**** | ****backend/pipeline/detectors/base.py**** reuse contract [verified] | No re-architecture of windowing/stitch. New code = native runner, USB/SD detection, app shell, packaging. |

40 | D6 | ****Feasibility is gated on P0.**** No app shell, no packaging work starts until the compute spike returns a verdict. | This doc; ****decision-rigor-norm**** (NUMBERS INFORM, FOUNDATIONS DECIDE) | P0 is the only un-gated deliverable; P1?P3 are owner-greenlit ****after**** P0 numbers. |

41
42 **## 3. Foundation ? research / prior art [design] / [researched]**

43
44 **### 3.1 The compute reality (the crux) ? what we know vs must measure**
45 - ****WASB on a discrete laptop GPU is acceptable**** [verified, measured]: on the owner's ****RTX 2070 Super Max-Q (8 GB)****, WASB inference is the pipeline driver at ****~3.2?3.4x real-time**** (?3.4 GPU-min per footage-min, ~0.055 s/frame), peak VRAM ~5.7?6.0 GB; proxy (NVENC) ~0.2x RT; 4K stitch (CPU libx264) ~16 s/rally; end-to-end ?5x clip length. Source: `current-state.md` `COST\_SCALING.md` checkpoint (6 Boxhill clips, 2026-06-18).

46 - ****The typical enthusiast laptop has NO discrete GPU.**** A 10-min 4K clip at 3.4x RT is ~34 GPU-min ****with**** a 2070S. On integrated graphics / CPU-only the wall-clock is unknown and

could be 10x+ worse ? possibly ****hours per clip****, which would kill the one-click promise.

47 - ****Apple Silicon has no CUDA.**** WASB/TrackNet weights are PyTorch; PyTorch MPS (Metal) backend ***may*** run them, but op-coverage and speed on the WASB model are ****unverified**** ``[design]``. CPU fallback on Apple is also unverified.

48 - ****Owned-model tie-in.**** ``OWNED_MODEL_IMPLEMENTATION_PLAN.md`` already states the on-device answer: ***train in Python ? **export across the boundary**** (ONNX / Core ML / ExecuTorch / GGUF) ? a thin device shell runs the artifact with zero Python*, with the standing constraint to ****train with export-clean ops****. A smaller, exported, quantized shuttle detector is a credible answer if the CPU/MPS numbers on the current WASB weights are bad. ****This makes P0 partly a referendum on whether the desktop product needs the owned-model track.****

49

50 **### 3.2 The de-WSL native runner is already designed-for ``[verified]``**

9. user (2026-06-19T17:18:21.767Z)

1 # Internationalization (i18n) ? Kannada-first, by design

2

3 ****Status:**** Design (P0 deliverable; phases P1+ are owner-gated, not started).

4 ****Owner decision required**** before any code lands ? see `[Open decisions](#open-decisions-for-the-owner)`.

5

6 ---

7

8 **## 1. Context ? why this, why now**

9

10 The PMF beachhead is ****Bangalore badminton academies**** `([PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md))`.

11 Many academy coaches, parents, and players are far more comfortable in ****Kannada**** than English.

12 A product that only speaks English adds friction exactly at the point of first contact. So the

13 product must speak ****Kannada first****, ****Hindi next****, and extend to other Indian languages

14 (Tamil, Telugu, Marathi, ?) ****without a refactor****.

15

16 The owner's directive is precise: make the product ****internationalizable _by design****. Adding a

17 language must be ****"add a reviewed catalog" ? not "go re-touch the code."*** The translations

18 themselves can wait; ****the foundation that makes them cheap to add is the actual deliverable.****

19

20 Today there is ****zero i18n**** in the repo (no i18n/locale/gettext/babel anywhere). Every

21 user-facing string is hardcoded English across four surfaces:

22

23 | Surface | Where | Examples (today) |

24 |---|---|---|

25 | ****UI chrome**** | ``backend/api/static/{index.html,app.js,style.css}`` (~40?50 strings) | ``"Validate & Index Video"`, `alert("Validation Failed! ?")`, `` ` ${count} Rally Segment${count} > 1 ? 's' : '' } Selected` `` `` |

26 | ****Backend coach-facing text**** | ``backend/main.py`` ``HTTPException(detail=?)``; ``backend/pipeline/validators/*`` ``message``; ``backend/reporting/spec.yaml`` ``customer_verdict`` + ``per-rule customer_message`` | ``"Video file not found at '?'`"; `"Video quality rejected: Focus check failed?"`; `"? Your highlights are ready."` |`

27 | ****Self-navigating NL queries**** (owner ask, future) | not built yet | ****show me all rallies > 15 shots where I won**** ? the ****query and its answer**** must work in the user's language |

28 | ****Coach-mode recommendations**** (owner ask, future) | not built yet | localized recommendation text, not English |

29

30 A React/Vite SPA is planned to replace the static UI (`[UI_ALPHA_EXECUTION_PLAN.md]`, ADR-009)

31 in a ****separate private repo `avidullu/khelsutra`****. The design below is therefore deliberately

32 ****framework-agnostic****: the catalog format and locale-negotiation contract are shared,

```

so the
33     static UI today and the SPA tomorrow consume the *same* translation source.
34
35     **Scope confirmed with owner:** UI + coach-facing backend text + (designed-for, built-
later)
36     NL/coach features. **Translations are Gemini-drafted, human-reviewed** ? this is
*inference*
37     use of a paid LLM, which the [licensing guardrail](#guardrail-alignment) explicitly
allows
38     (it is *not* a training-data source). The owner reviews the first draft of every string.
39
40     ---
41
42     ## 2. Core architecture
43
44     ### 2.1 The "by design" invariant ? no hardcoded user-facing strings
45
46     Every coach-facing string is a **stable key** resolved at render time against the active
locale.
47     Code references `t("compile.button")`, never the English prose. This is the single rule
that
48     makes "add a language = add a catalog" true, and it is **enforced by CI** ($2.6) so it
cannot
49     silently regress as the codebase grows.
50

```

10. user (2026-06-19T17:18:22.692Z)

```

1     # Experiment harness ? A/B + shadow eval for rally-detection quality (no-regression by
design)
2
3     **Status:** **P1 IMPLEMENTED** (`backend/eval/experiment.py` +
`tests/test_experiment.py`, shipped
4     2026-06-13) ? the `Variant` layer, `compare()` (labeled A/B, LOVO-honest),
`compare_unlabeled()`
5     (SxS on NEW unlabeled footage), and the leaderboard. **P2?P4 remain owner-gated.**
Phases below are
6     sequenced but each is a separate PR.
7
8     > **Companion docs:** the cues this experiments *on* are in
9     > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); the scoring mechanics are
10    > [EVALUATION.md](EVALUATION.md); the no-regression gate + golden labels +
`baseline.json` come from
11    > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md); the modular-cue architecture is
**ADR-007**
12    > ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the distilled
classifier is
13    > **ADR-004**. The 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
14
15    ---
16
17    ## 1. Context ? why this, why now
18
19    PR #112 (merged) added the multi-signal fusion design ? shuttle + person + audio cues,
all **default
20    OFF**. The owner's directive for *how* we evolve quality from here:
21
22    > "Evolve the codebase so we can keep finding new ways to improve quality and
experiment with them ?
23    > or add them as a signal ? without suffering any regressions if the idea doesn't work
out. Rolling
24    > back the code would be too dangerous."
25
26    So **experimentation and deployment must be decoupled.** The contract:

```

```

27
28     1. Merge experimental code freely ? it's **gated OFF**, so it cannot change production
behavior.
29     2. Test it **offline against golden labels** (most experiments never touch the served
path at all).
30     3. Promote a variant to default **only on measured lift**, through a CI eval gate.
31     4. **"Rollback" = flip a config flag, never `git revert`. The proven path is a pinned
variant that is
32         always present.
33
34     **The pleasant surprise (Explore audit, 2026-06-13):** ~80% of this harness already
exists and is
35     simply not composed *as a system*. The missing piece is a thin **variant/experiment
layer + the
36     discipline**, not a platform. S4 is the exact code surface so the next session does not
re-discover it.
37
38     ---
39
40     ## 2. The thesis ? separate expensive DETECTION from cheap DECISION
41
42     The expensive, GPU/WSL-bound, risky part is **detection**: "what signals exist per
frame" (shuttle
43     WASB trajectory, person tracks/boxes, audio hits). The cheap, swappable, pure-Python
part is the
44     **decision**: "given those signals, where are the rallies" (windowing + gates + the
classifier).
45
46     > **Run detection once per video, persist all per-cue signals + per-candidate features,
then re-score
47     > any number of variants OFFLINE ? deterministically, with no GPU and zero production
risk.**
48
49     This is not aspirational ? `distill_local.py` and `calibrate_local.py` already do
exactly this on
50     stored WASB trajectories. The candidate-segments table (`stats` JSON) is the persistence
substrate.
51     ? Most experiments are pure re-scoring of stored data; they never enter the served
pipeline.
52
53     ---
54
55     ## 3. What already exists (the audit ? do NOT rebuild)

```

11. user (2026-06-19T17:18:23.040Z)

```

1     # Ablation layer (Phase 0) ? per-signal marginal contribution
2
3     A *systematic, repeatable* version of the earlier one-off ablation:
4     `backend/eval/ablation.py` runs it and it EXCLUDES one signal, measures whether quality
dropped,
5     and reports each signal's **marginal ?F1** (full set vs full-minus-that-signal) with a
bootstrap
6     95% CI + paired sign test + the C4 over-segmentation ratio. The broader design (the
ledger time
7     series, interaction handling, per-user ablation, the phased plan) is not yet split into
its own
8     doc; this note covers only the shipped Phase-0 runner.
9
10    ## What it is (and is not)
11
12    **A pure driver.** No new scoring/stats math ? every number defers to the shipped
13    `experiment.compare` + `eval_stats` engine, exactly as `compare` was a driver over
14    `metrics`/`calibration`. For a signal `s`:
15

```

```

16     ``
17     ?(s) = score(full set S) ? score(S \ {s})          # leave-one-out drop, ?F1 primary
18     ``
19
20     is `experiment.compare(videos, variant_a=without_s, variant_b=full)` ? A is the ablated
(smaller)
21     variant, B is full, so the reported B?A ?F1 IS ?(s): a per-video-paired ?F1 with
bootstrap CI
22     + exact sign test, LOVO-honest for learned variants.
23
24     A signal is either a feature column (person/audio/shuttle columns the learned scorer
weights,
25     ablated by dropping the column) or a gate knob (Gate-A's `min_crossings` threshold,
ablated by
26     zeroing the knob). The runner handles both uniformly.
27
28     ## Two granularities
29
30     - LOSO (`--granularity signal`) ? leave one column-set out at a time (fine-grained).
31     - LOGO (`--granularity group`) ? leave a whole producer's columns out at once (the
owner's
32     "is this old SIGNAL still relevant?" question at cue granularity). Default.
33
34     ## The honest small-N contract
35
36     Inherited from `eval_stats` / `per_user_gate` (no new statistics):
37
38     - Never a bare mean ? every row ships a CI + the exact sign test.
39     - A signal earns its keep only when removal drops F1 with the ?F1 CI excluding 0
AND the
40     sign test agreeing (the `per_user_gate` bar, in reverse).
41     - Structural guards are exempt from auto-demotion. Gate-A (the held-shuttle false-
positive
42     killer) is marked `structural`; it is reported honestly but verdict is always
43     `structural-protected`, never flagged dead weight on "no measured lift" ? its value is
in
44     failure modes the golden corpus may not contain.
45     - The verdict vocabulary is closed and never over-claims: `clears-bar` / `suggestive` /
`inert`
46     / `structural-protected`. One wash at n=6 is `inert` (a candidate), never
"irrelevant".
47
48     ## Run it
49
50     ```bash
51     # LOGO contribution table over the learned full set (shuttle+gate_a+person+audio):
52     python -m backend.eval.ablation --features-dir output [--granularity signal] \
53         [--pdf ablation.pdf] [--json ablation.json]
54
55     # Reproduce the hand-measured Gate-A result (the framework's litmus):

```

12. assistant (2026-06-19T17:18:26.554Z)

Now let me gather cross-reference evidence: check for already-archived copies and citations across CLAUDE.md, NEXT_STEPS.md, README.md, and the whole repo.

13. user (2026-06-19T17:18:29.139Z)

```

=== archived stems ===
2026-06-first-ab-experiment
=== END ===

```

14. user (2026-06-19T17:18:30.895Z)

CLAUDE.md:1
 docs\ANALYZERS_AND_RECONCILERS.md:7
 tests\test_async_jobs.py:1
 tests\test_database.py:2
 backend\eval\ablation.py:1
 docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md:1
 docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:6
 backend\eval\experiment.py:9
 backend\eval\per_user_eval.py:1
 backend\eval\eval_stats.py:1
 backend\eval\per_user_regression.py:2
 backend\eval\score_calibration.py:1
 backend\eval\segmentation_metrics.py:1
 docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:4
 docs\NEXT_STEPS.md:2
 docs\MULTI_SIGNAL_FUSION_PLAN.md:1
 docs\QUALITY_ITERATIONS.md:4
 docs\PROMINENT_COURT_DETECTION.md:2
 backend\infrastructure\database.py:3
 docs\COMMERCIALIZATION.md:1
 docs\EXPERIMENT_HARNESS.md:1
 docs\CODE_MAP.md:4
 docs\I18N_PLAN.md:1
 docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:2
 docs\DESKTOP_APP_PLAN.md:1
 docs\README.md:5
 docs\GOLDEN_REGRESSION_FIXTURES.md:2
 docs\PLATFORM_ARCHITECTURE.md:1
 docs\GOLDEN_SET_VENDORING.md:1
 docs\RALLY_DETECTION_GAP_CLOSURE.md:2
 docs\PERSONALIZATION_PLAN.md:2
 docs\RALLY_QUALITY_RESEARCH.md:7
 tests\test_experiment.py:1
 backend\pipeline\personalization__init__.py:1
 backend\pipeline\personalization\serving.py:2
 backend\pipeline\segmenters\fusion_hybrid.py:2

Found 86 total occurrences across 36 files.

15. assistant (2026-06-19T17:18:34.671Z)

No archived copies exist for any of these seven docs. Now let me check the specific citations in the authoritative live sources (CLAUDE.md, NEXT_STEPS.md, README.md) and look at the progress-tracker / definition-of-done sections for the docs that claim partial implementation.

16. user (2026-06-19T17:18:37.889Z)

```

=== CLAUDE.md ===
19: "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md` (A/B
+ shadow
=== NEXT_STEPS.md ===
60: `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
68:- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**;
```

promote to default **only on measured LOVO lift** through `run\_regression.py` (exit 0/1/3);
 pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

```

=== README.md ===
2:- [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? **? Foundation doc** for closing
the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
**adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
falsifiable success criteria**, and ($7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
```

(project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.

7:- EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

9:- ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

10:- PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization** feature on a generalization-first baseline: a thin per-user **tuning** layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; **"a tool that gets better the more you help it"**). Records the **locked owner decisions** (identity, storage, **min_n** semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

15:- DESKTOP_APP_PLAN.md ? **Tracked scoping doc** (2026-06-18, owner-gated, **[design]** ? **NOTHING** built): **package the local pipeline as a cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical **no-GPU** enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike** ? P1 native detector ? P2 app shell ? P3 packaging/installers. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **LocalDesktop ComputeBackend** of PLATFORM_ARCHITECTURE.md \$3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

17. assistant (2026-06-19T17:18:41.794Z)

All seven docs are indexed as LIVE/design entries in docs/README.md, and EXPERIMENT_HARNESS is cited as a live next-source in both CLAUDE.md and NEXT_STEPS.md. Let me check the I18N and PER_VIDEO references in README, and read the progress trackers / definition-of-done sections to confirm none are terminal (DONE/COMPLETED).

18. user (2026-06-19T17:18:44.388Z)

```
=== README I18N/PER_VIDEO ===
=== ABLATION in README ===
```

19. user (2026-06-19T17:18:47.302Z)

```
===== ANALYZERS_AND_RECONCILERS =====
3:**Status:** **DESIGN ? owner-gated, NOT started.** A peer to
396:## 7. No-regression discipline (identical to the harness contract)
414:## 8. How it composes with fusion P2
===== PERSONALIZATION_PLAN =====
3:**Status: DESIGN ? owner-adopted (hybrid); Phase-0 engineering landing.** This is the
authoritative
150:## 7. References
===== PER_VIDEO_OUTPUT_LAYOUT =====
3:**Status:** **DESIGN ? owner-gated, NOT started.** Proposed convention so processing N videos
===== DESKTOP_APP_PLAN =====
```

```

3:> **Status:** `DRAFT` ? `owner-gated` (scoping doc only; **nothing is built**) . **Owner:**
avidullu . **Created:** 2026-06-18 . **Last updated:** 2026-06-18
5:> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once the owner
greenlights P0).
124:## 7. Deliverables & progress tracker ? **source of truth**
135:## 8. Open questions ? owner / external
147:## 9. Definition of done
===== I18N_PLAN =====
3:**Status:** Design (P0 deliverable; phases P1+ are owner-gated, not started).
===== EXPERIMENT_HARNESS =====
3:**Status:** **P1 IMPLEMENTED** (`backend/eval/experiment.py` + `tests/test_experiment.py`,
shipped
173:## 7. What to build later (gap list ? thin, additive, owner-gated)
192:## 8. Phasing (owner-gated; ONE PR per slice, off `master`)
===== ABLATION_LAYER =====

```

20. assistant (2026-06-19T17:18:51.260Z)

All status headers confirm these are DESIGN/in-progress/partially-implemented docs ? none are terminal. Note that I18N_PLAN, PER_VIDEO_OUTPUT_LAYOUT, and ABLATION_LAYER are NOT indexed in README.md (the grep returned nothing for them there), but they are referenced elsewhere. Let me confirm the inbound referrers for the three not-in-README docs and check DESKTOP_APP_PLAN's progress tracker to be thorough.

21. user (2026-06-19T17:18:56.503Z)

```

124    ## 7. Deliverables & progress tracker ? **source of truth**
125    Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
**Everything below is owner-gated; only this doc (the scoping PR) is in flight today.**
126
127    | ID | Deliverable | Depends on | Gated? | Status | PR |
128    |----|-----|-----|-----|-----|----|
129    | **D0** | **This scoping doc** (`DESKTOP_APP_PLAN.md` + README index) | ? | No | ? | |
*(this PR)* |
130    | **P0** | **Compute-feasibility spike (make-or-break).** Measure WASB throughput for a
~10-min clip on: (a) CPU-only (torch-CPU & onnxruntime), (b) Apple MPS, (c) confirm CUDA-native
(no WSL). Output: a wall-clock table + a go/no-go verdict + "do we need a smaller model?"
recommendation. **No app code.** | D0 | **Owner greenlight** | ? | ? | |
131    | **P1** | **De-WSL native detector port** (`NativeWasbRunner`, per-OS): CUDA-native
(Win/Linux), MPS/CPU fallback, identical `Frame,Visibility,X,Y` CSV; reuse windowing verbatim;
healthcheck per backend. | P0 verdict | Yes | ? | ? | |
132    | **P2** | **Desktop app shell + one-click flow**: USB/SD detect ? list ? proxy ? native
detect ? stitch ? export; shell choice (CLI MVP ? Tauri) confirmed; "no upload / no original-
copy" enforced & testable. | P1 | Yes | ? | ? | |
133    | **P3** | **Packaging + code-signing installers** per OS (Win MSIX/Authenticode, mac
.dmg + Developer-ID notarization, Linux AppImage/Flatpak); LGPL ffmpeg bundling; MIT/Apache/BSD
audit of the bundle. | P2 | Yes | ? | ? | |
134
135    ## 8. Open questions ? owner / external
136    **Blocking gates (owner decides before the dependent phase):**
137    1. **(gates P1?P3) Greenlight P0?

```

```

144     6. Auto-update strategy (Tauri updater / Sparkle / none for the pilot).
145     7. Does the desktop app reuse the optional Gemini toggle at all, or stay strictly local
for v1?
146
147     ## 9. Definition of done
148     This subproject is DONE (? flip Status, archive) when:
149     - [ ] **P0** produced a measured throughput table (CPU / MPS / CUDA-native) and an
owner-accepted go/no-go verdict (incl. whether a smaller model is required). *(P0 alone is a
valid stopping point if the verdict is "not feasible without the owned model" ? then this doc
hands off to `OWNED_MODEL_IMPLEMENTATION_PLAN`.)*
150     - [ ] **P1** the native (de-WSL) detector runs on ?1 GPU OS and ?1 CPU/MPS path,
producing byte-compatible trajectories (regression vs the WSL runner on a shared clip).
151     - [ ] **P2** a non-technical user can insert a card and get a reel with no terminal, no
upload, and no copied original ? verified end-to-end on ?1 OS.
152     - [ ] **P3** signed/notarized installers exist for the greenlit OSes; the bundle passes
an MIT/Apache/BSD (+ LGPL-ffmpeg) license audit.
153     - [ ] $7 rows all ? and the headline feasibility verdict is recorded for posterity.
154
155     ## 10. References
156     **Internal code anchors `[verified]`:**
[tools/generate_highlights.py](../tools/generate_highlights.py) (local orchestration; `--skip-
ai-handoff`, `--wasb-stream`, `--detect-from`, `_WSL_DETECTORS`, `_atomic_publish`) .
[backend/pipeline/detectors/base.py](../backend/pipeline/detectors/base.py) (`DetectorRunner`
ABC + the documented Linux-native de-risk path + `WslCommandMixin`) .
[backend/pipeline/detectors/wasb_runner.py](../backend/pipeline/detectors/wasb_runner.py) (the
WSL adapter to replace) .
[backend/pipeline/detectors/wasb_infer.py](../backend/pipeline/detectors/wasb_infer.py) (the
torch inference core to lift native) . [backend/pipeline/detectors/tracknet_runner.py](../back
end/pipeline/detectors/tracknet_runner.py) (shared `trajectory_to_action_windows`) .
[backend/pipeline/segmenters/wasb_hybrid.py](../backend/pipeline/segmenters/wasb_hybrid.py) .
`backend/pipeline/stitcher/compiler.py::VideoCompiler` . `pyproject.toml` + `indexer` console
entry (#167).
157     **Internal docs:** [VIDEO_LOCALITY_MODEL.md](VIDEO_LOCALITY_MODEL.md) (L0 fully-local
tier; Q05 packaging) . [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) §3.2
(ComputeBackend seam) .
[OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) (export-clean on-
device model) . [COMMERCIALIZATION.md](COMMERCIALIZATION.md) (C4 paid-LLM no-train;
MIT/Apache/BSD-only) . [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) (#7 packaging,
DONE) . [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md) (the WSL setup being superseded for
shipped builds).
158     **Measured source:** `current-state.md` 2026-06-18 `COST_SCALING.md` checkpoint (WASB
~3.4x RT on RTX 2070 Super Max-Q, peak VRAM ~5.7?6.0 GB).

```

22. user (2026-06-19T17:18:56.783Z)

```

=== I18N_PLAN referrers ===
docs/COMMERCIALIZATION.md
docs/I18N_PLAN.md
docs/PLATFORM_ARCHITECTURE.md
=== PER_VIDEO referrers ===
docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs/CODE_MAP.md
docs/QUALITY_ITERATIONS.md
=== ABLATION_LAYER referrers ===
docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs/RALLY_QUALITY_RESEARCH.md
backend/eval/ablation.py
backend/eval/___pycache___/ablation.cpython-313.pyc
=== ANALYZERS referrers (files) ===
docs/archives/quality-iterations/2026-06-first-ab-experiment/README.md
docs/EXPERIMENT_HARNESS.md
docs/QUALITY_ITERATIONS.md
docs/RALLY_QUALITY_RESEARCH.md
docs/README.md

```



```

backend/eval/eval_stats.py
backend/eval/experiment.py
backend/eval/per_user_eval.py
backend/eval/per_user_regression.py
backend/eval/score_calibration.py
backend/eval/segmentation_metrics.py
backend/eval/___pycache___/eval_stats.cpython-313.pyc
backend/eval/___pycache___/experiment.cpython-313.pyc
backend/eval/___pycache___/per_user_eval.cpython-313.pyc
backend/eval/___pycache___/per_user_regression.cpython-313.pyc
backend/eval/___pycache___/score_calibration.cpython-313.pyc
backend/eval/___pycache___/segmentation_metrics.cpython-313.pyc
=== PERSONALIZATION referrers ===
docs/CODE_MAP.md
docs/QUALITY_ITERATIONS.md
docs/README.md
backend/infrastructure/database.py
backend/pipeline/personalization/serving.py
backend/pipeline/personalization/___init___py
backend/pipeline/personalization/___pycache___/serving.cpython-313.pyc
backend/pipeline/personalization/___pycache___/___init___py
=== DESKTOP referrers ===
docs/DESKTOP_APP_PLAN.md
docs/README.md

```

23. assistant (2026-06-19T17:19:06.377Z)

I have all the evidence I need. Every doc in this batch is a LIVE design/plan or in-progress/partially-implemented doc, none is in a terminal (DONE/COMPLETED/REJECTED/superseded) state, none has an already-archived copy, and all are cited as live/design sources. All seven are KEEP_LIVE.

Summary of evidence:

- ****No archived copies exist**** for any of the seven (Glob of docs/archives confirmed only `2026-06-first-ab-experiment/` matches, which is a quality-iterations checkpoint, not a copy of these docs).
- ****ANALYZERS_AND_RECONCILERS, PER_VIDEO_OUTPUT_LAYOUT**** ? status "DESIGN ? owner-gated, NOT started" = LIVE future work.
- ****PERSONALIZATION_PLAN**** ? "DESIGN ? owner-adopted; Phase-0 landing", #141/#142 shipped = in-progress.
- ****DESKTOP_APP_PLAN**** ? "DRAFT, owner-gated, nothing built", \$7 tracker + \$9 DoD all unchecked = LIVE tracked scoping doc.
- ****I18N_PLAN**** ? "Design (P0 deliverable; P1+ owner-gated)" = LIVE.
- ****EXPERIMENT_HARNESS**** ? "P1 IMPLEMENTED; P2?P4 owner-gated"; cited as a LIVE next-source in CLAUDE.md L19 and NEXT_STEPS.md L60/L68 = LIVE (and would be ASK_OWNER even if it claimed done).
- ****ABLATION_LAYER**** ? describes the shipped Phase-0 runner `backend/eval/ablation.py`; living reference for active code = LIVE.

24. user (2026-06-19T17:19:41.206Z)

Structured output provided successfully

25. assistant (2026-06-19T17:19:52.013Z)

All seven docs in my batch are KEEP_LIVE. None is in a terminal state and none has an already-archived copy.

Key findings:

- ****None of the seven has an archived copy**** ? Glob of `docs/archives/` and its subdirs returned no matching stems (the only near-match, `2026-06-first-ab-experiment/`, is a quality-iterations checkpoint, not a copy of any batch doc).
- ****All are LIVE design/in-progress/reference docs:****
 - `ANALYZERS\_AND\_RECONCILERS.md`, `PER\_VIDEO\_OUTPUT\_LAYOUT.md` ? "DESIGN ? owner-gated, NOT started" (owner-gated future work = LIVE, not terminal).

- `PERSONALIZATION\_PLAN.md` ? authoritative, "Phase-0 landing", #141/#142 shipped, more pending (in-progress).
- `DESKTOP\_APP\_PLAN.md` ? fresh (2026-06-18) DRAFT scoping doc; §7 tracker all gated, §9 DoD fully unchecked; its own lifecycle line says archive only when DONE.
- `I18N\_PLAN.md` ? "Design (P0 deliverable; P1+ owner-gated)"; zero i18n built yet.
- `EXPERIMENT\_HARNESS.md` ? only P1 shipped, P2?P4 pending; additionally cited as a LIVE next-source in `C:\Users\avidu\Projects\badminton-highlight-indexer\CLAUDE.md` (line 19) and `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT\_STEPS.md` (lines 60, 68).
- `ABLATION\_LAYER.md` ? living reference for the shipped, in-use runner
`C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\ablation.py`.

No ARCHIVE_NOW, DEDUPE_STUB, or ASK_OWNER verdicts in this batch ? nothing is safe to move to `docs/archives/`.